

Polar coordinates

Basic skills

Q1. a) P: $(x, y) = (3, 0)$

$$r^2 = 3^2 + 0^2 \quad \Theta = \tan^{-1}\left(\frac{0}{3}\right) = 0$$

Use $r^2 = x^2 + y^2$,
 $\tan \theta = \frac{y}{x}$

$$r^2 = 9 \Rightarrow r = 3 \quad \Theta = 0 \quad \text{so } (r, \theta) = (3, 0)$$

b) Q: $(x, y) = (0, 3)$

$$r^2 = 0^2 + 3^2 \quad \Theta = \frac{\pi}{2}$$

$$r^2 = 9 \Rightarrow r = 3$$

cannot use
 $\tan \theta = \frac{y}{x}$ as $x = 0$
(on y-axis $\frac{\pi}{2}, \frac{3\pi}{2}, -\frac{\pi}{2}$
by inspection)

$$\text{so } (r, \theta) = (3, \frac{\pi}{2})$$

c) R: $(x, y) = (0, -7)$

$$r^2 = 0^2 + (-7)^2 \quad \Theta = -\frac{\pi}{2}$$

or $\frac{3\pi}{2}$

$$r = 7 \quad \text{so } (r, \theta) = (7, -\frac{\pi}{2})$$

d) S: $(x, y) = (4, 4)$

$$r^2 = 4^2 + 4^2$$

$$\Theta = \tan^{-1}\left(\frac{4}{4}\right) = \frac{\pi}{4}$$

remember r in
rads

$$r^2 = 32$$

$$r = \sqrt{32} = 4\sqrt{2}$$

$$\text{so } (r, \theta) = (4\sqrt{2}, \frac{\pi}{4})$$

e) $t: (x, y) = (-2, -3)$

-ve be careful

$$r^2 = (-2)^2 + (-3)^2$$

$$r^2 = 4 + 9 = 13$$

$$r = \sqrt{13}$$

$$\Theta = \tan^{-1}\left(\frac{-3}{-2}\right) = 0.983$$

but as both -ve
need

$$\Theta = -\pi + 0.983$$

So $(r, \Theta) = (\sqrt{13}, 0.983 - \pi)$ (or $\pi + 0.983$)

Competency skills

Q2. a) $(x, y) = (3, 4)$

$$r^2 = 3^2 + 4^2 = 25$$

$$r = 5$$

$$\Theta = \tan^{-1}\left(\frac{4}{3}\right) = 0.927$$

$$(r, \Theta) = (5, 0.927)$$

b) $(x, y) = (-5, 12)$

$$r^2 = (-5)^2 + 12^2 = 169$$

$$r = 13$$

$$\Theta = \tan^{-1}\left(\frac{12}{-5}\right) = -1.18$$

$$(r, \Theta) = (13, -1.18)$$

c) $(x, y) = (-3, -3)$

$$r^2 = (-3)^2 + (-3)^2 = 18$$

$$r = 3\sqrt{2}$$

$$\Theta = \tan^{-1}\left(\frac{-3}{-3}\right) = \frac{\pi}{4}$$

however two -ve's

so $\frac{\pi}{4} - \pi = -\frac{3\pi}{4}$

$$(r, \Theta) = (3\sqrt{2}, -\frac{3\pi}{4})$$

$$d) (x, y) = (5, -1)$$

$$\Theta = \tan^{-1}\left(\frac{-1}{5}\right) = -0.197$$

$$r^2 = 5^2 + (-1)^2 = 26$$

$$r = \sqrt{26}$$

$$(r, \Theta) = (\sqrt{26}, -0.197)$$

$$Q3. a) (r, \theta) = (\sqrt{2}, \frac{\pi}{4})$$

$$x = r \cos \theta = \sqrt{2} \cos \frac{\pi}{4} = 1$$

$$y = r \sin \theta = \sqrt{2} \sin \frac{\pi}{4} = 1$$

$$(x, y) = (1, 1)$$

$$b) (r, \theta) = (\sqrt{2}, \frac{3\pi}{4})$$

$$x = r \cos \theta = \sqrt{2} \cos \left(\frac{3\pi}{4}\right) = -1$$

$$y = r \sin \theta = \sqrt{2} \sin \left(\frac{3\pi}{4}\right) = 1$$

$$(x, y) = (-1, 1)$$

$$c) (r, \theta) = (3, \frac{7\pi}{6})$$

$$x = r \cos \theta = 3 \cos \left(\frac{7\pi}{6}\right) = -\frac{3\sqrt{3}}{2}$$

$$y = r \sin \theta = 3 \sin \left(\frac{7\pi}{6}\right) = -\frac{3}{2}$$

$$(x, y) = \left(-\frac{3\sqrt{3}}{2}, -\frac{3}{2}\right)$$

$$d) (r, \theta) = (7, \frac{10\pi}{6})$$

$$x = r \cos \theta = 7 \cos \left(\frac{10\pi}{6}\right) = \frac{7}{2}$$

$$y = r \sin \theta = 7 \sin \left(\frac{10\pi}{6}\right) = -\frac{7\sqrt{3}}{2}$$

$$(x, y) = \left(\frac{7}{2}, -\frac{7\sqrt{3}}{2}\right)$$

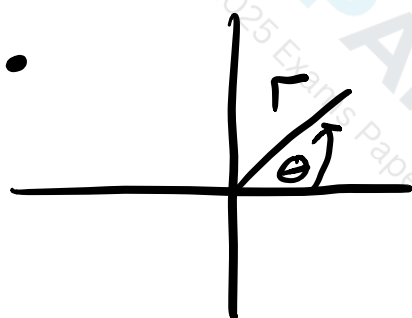
Q4. $\sqrt{x^2+y^2} = \sqrt{r^2} = r$
 $\searrow$
 $r^2 = x^2 + y^2$

Q5. $r = \frac{s}{\sin \theta} \Rightarrow s = r \sin \theta$

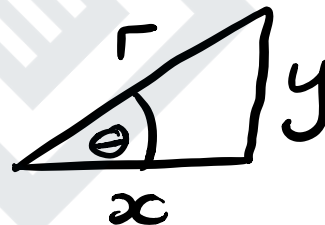
As $r \sin \theta = y$ we have $y = s$

Advanced skills

Q6.



triangle

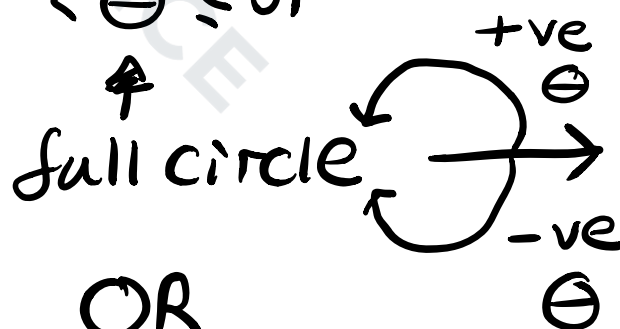


So by trig $\cos \theta = \frac{x}{r}$

So $x = r \cos \theta$

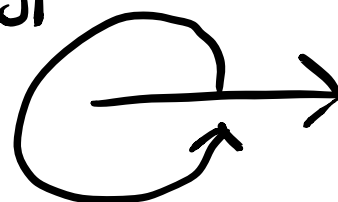
Q7. $r \geq 0$ and $-\pi < \theta \leq \pi$

distance
from base point
i.e. (0,0)



OR

$0 \leq \theta < 2\pi$



Q8. $y^2 = 4(1-x)$

$$(r \sin \theta)^2 = 4(1 - r \cos \theta)$$

$$r^2 \sin^2 \theta = 4 - 4r \cos \theta$$

$$r^2 \sin^2 \theta + 4r \cos \theta - 4 = 0 \quad \text{solve quadratic for } r$$

$$r = \frac{-4 \cos \theta \pm \sqrt{(4 \cos \theta)^2 - 4 \times \sin^2 \theta \times -4}}{2 \sin^2 \theta}$$

$$= \frac{-4 \cos \theta \pm \sqrt{16(\cos^2 \theta + \sin^2 \theta)}}{2 \sin^2 \theta}$$

$$= \frac{-4 \cos \theta \pm 4}{2 \sin^2 \theta} = \frac{-2 \cos \theta \pm 2}{\sin^2 \theta}$$

Either $r = \frac{-2 \cos \theta + 2}{\sin^2 \theta}$ or $r = \frac{-2(1 + \cos \theta)}{\sin^2 \theta}$

Q9. $r = \frac{3}{\sin\theta - 2\cos\theta}$

$$r \sin\theta - 2r \cos\theta = 3$$

$$y - 2x = 3$$

$y = 3 + 2x$ intersects with $x + y = 6$

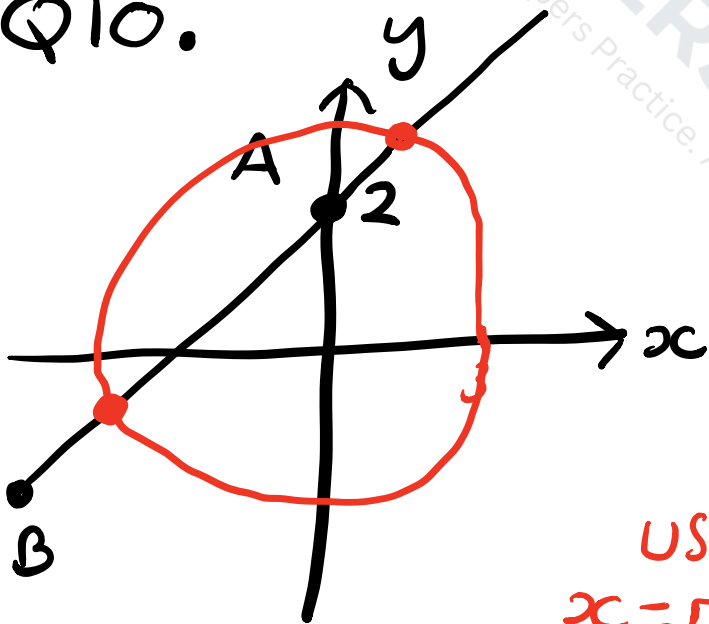
at:

$$x + 3 + 2x = 6 \Rightarrow x = 1$$

$$\Rightarrow y = 5$$

so $(1, 5)$

Q10.



$$A: (r, \theta) = (2, \frac{\pi}{2})$$

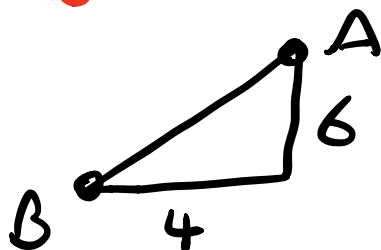
$$(x, y) = (0, 2)$$

$$B: (r, \theta) = (4\sqrt{2}, \frac{5\pi}{4})$$

$$(x, y) = (-4, -4)$$

use $x = r \cos\theta$
 $y = r \sin\theta$

Line AB:



so gradient

$$m = \frac{6}{4} = \frac{3}{2}$$

so $y = \frac{3}{2}x + c$ sub (0,2)

$$2 = c \Rightarrow y = \frac{3}{2}x + 2$$

intersect with $x^2 + y^2 = 3^2 = 9$

$$\Rightarrow x^2 + \left(\frac{3}{2}x + 2\right)^2 = 9$$

$$\Rightarrow x^2 + \frac{9}{4}x^2 + 6x + 4 = 9$$

$$\Rightarrow \frac{13}{4}x^2 + 6x - 5 = 0$$

$$\Rightarrow 13x^2 + 24x - 20 = 0$$

$$\Rightarrow x = \frac{-24 \pm \sqrt{24^2 - 4 \times 13 \times -20}}{2 \times 13}$$

$$= \frac{-12 \pm 2\sqrt{101}}{13}$$

so $y = \frac{3}{2}x \left(\frac{-12 \pm 2\sqrt{101}}{13} \right) + 2$

$$= \frac{8 \pm 3\sqrt{101}}{13}$$